

SEVERING THE TETHER THE TETHER

The Architecture of Onboard AI

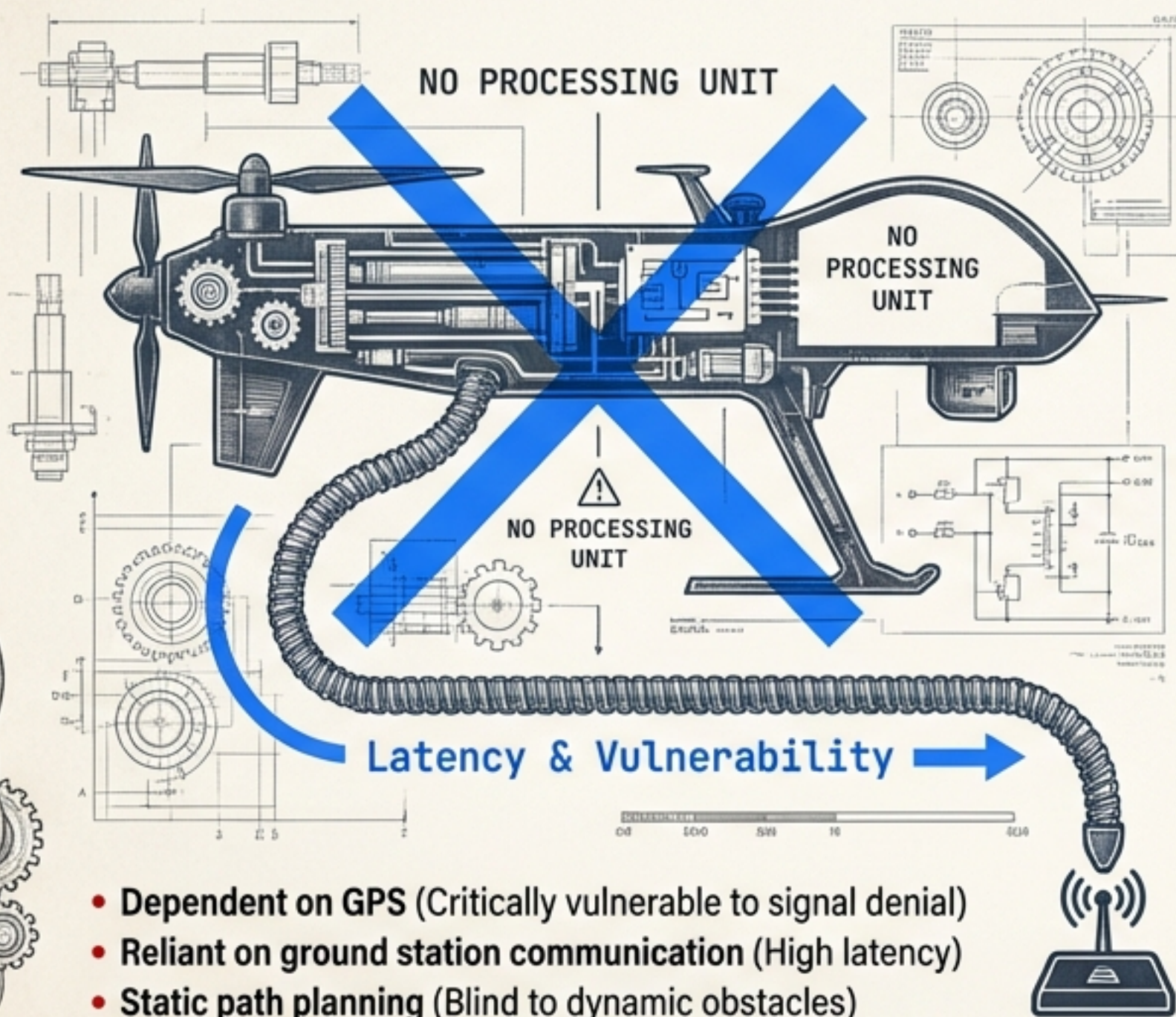


From Remote-Controlled Machines
to Self-Governing Organisms.

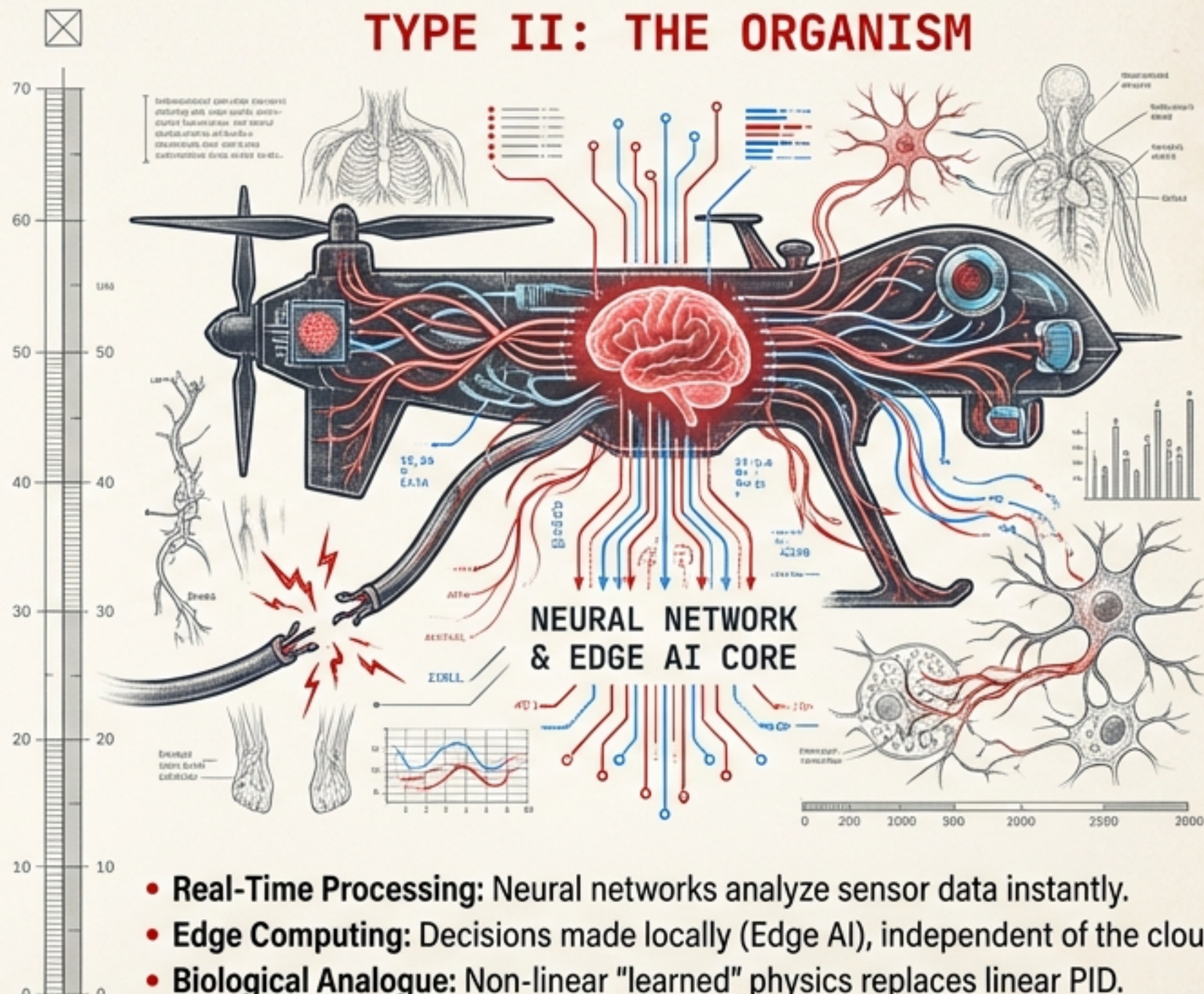
A Synthesis of Perception, Control,
Compute, and Swarm Intelligence.

The Imperative for Onboard Intelligence

TYPE I: THE PUPPET



TYPE II: THE ORGANISM



To fly alone, a drone must possess the computational equivalent of eyes, a brain, and a nervous system.

The Anatomy of Autonomy



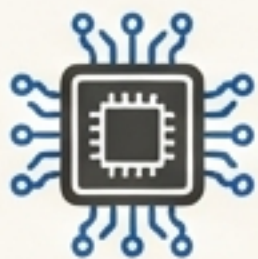
THE EYES (Perception)

Lightweight Object
Detection (UAD-YOLOv8)



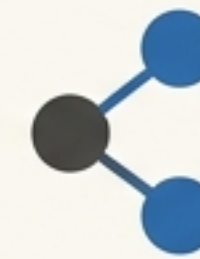
THE BRAIN (Control)

Deep Reinforcement
Learning (SAC)



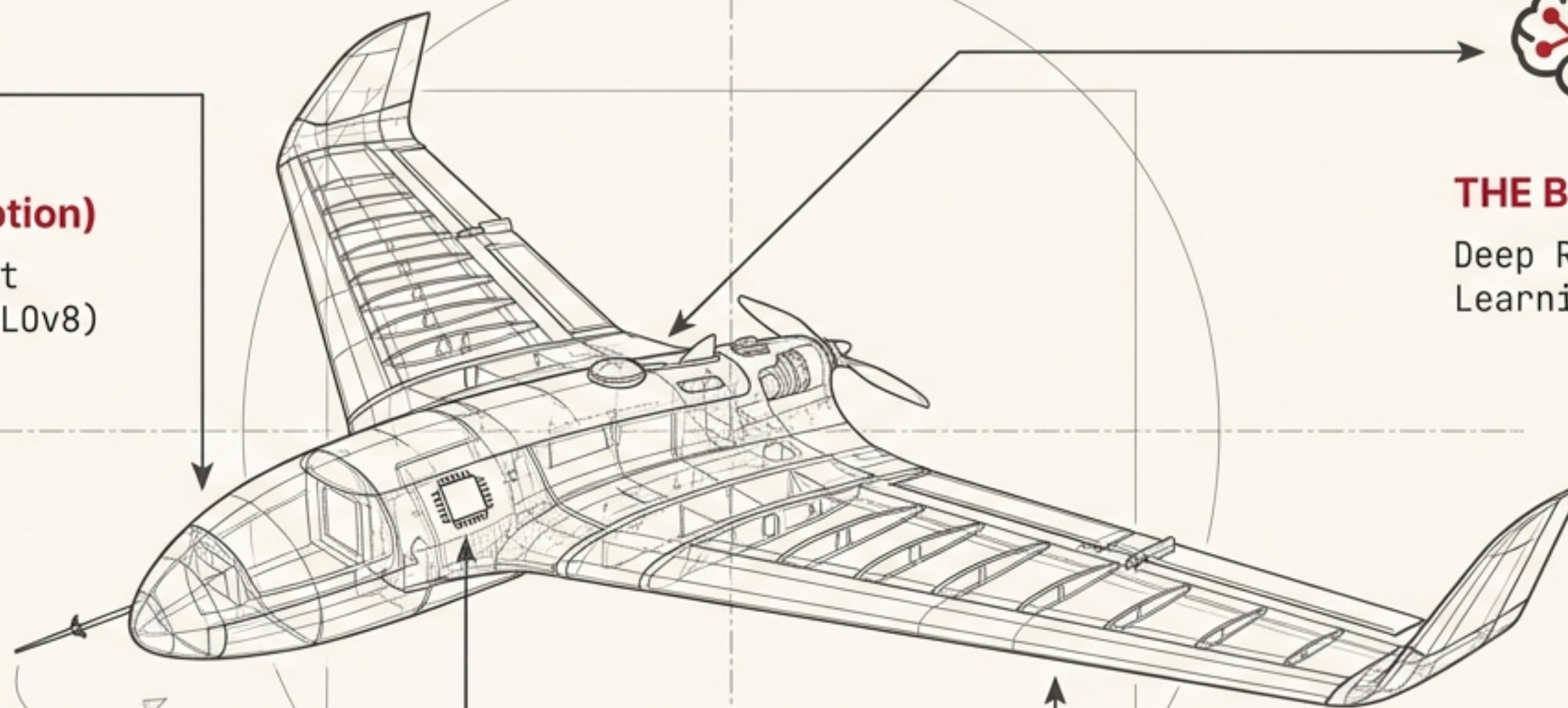
THE CORTEX (Hardware)

Edge-AI Accelerators
(Jetson vs Coral)



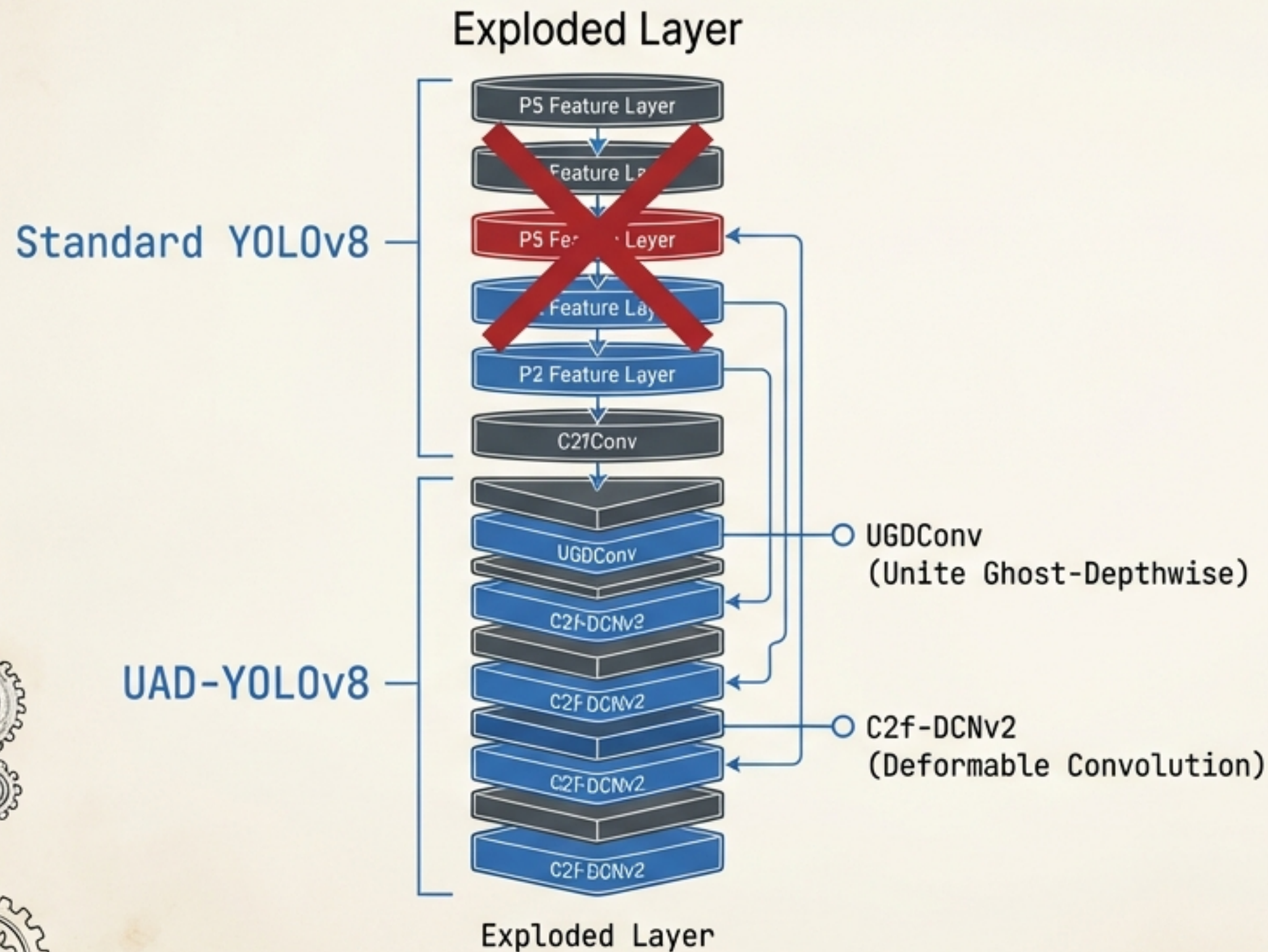
THE SOCIETY (Swarm)

Decentralized Cohesion



The Eyes: Optimizing Perception

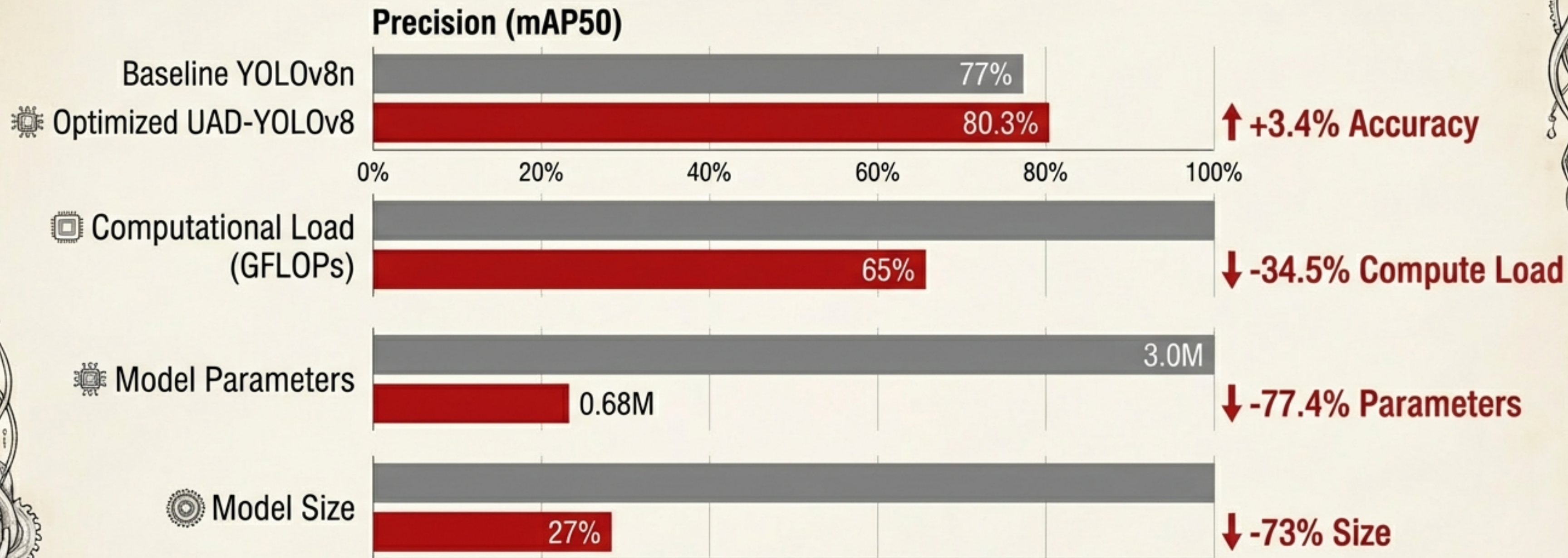
UAD-YOLOv8 — Stripping the Fat for Speed.



Standard Computer Vision is too metabolically expensive for small UAVs. We require an algorithm that prioritizes small targets and sheds unnecessary features.

- **P5 Layer Removal:** Deleting the feature layer used for large objects optimizes the network specifically for small, distant aerial obstacles.
- **UGDConv Integration:** Replaces standard convolutions to drastically reduce parameter count.
- **C2f-DCNv2:** Deformable convolutions handle irregular obstacle shapes better than static convolutions.

Perception Metrics: Doing More with Less



The UAD-YOLOv8 architecture achieves the **Edge AI "holy grail"**: increasing detection **accuracy** for small targets while **slashing computational cost** by nearly three-quarters.

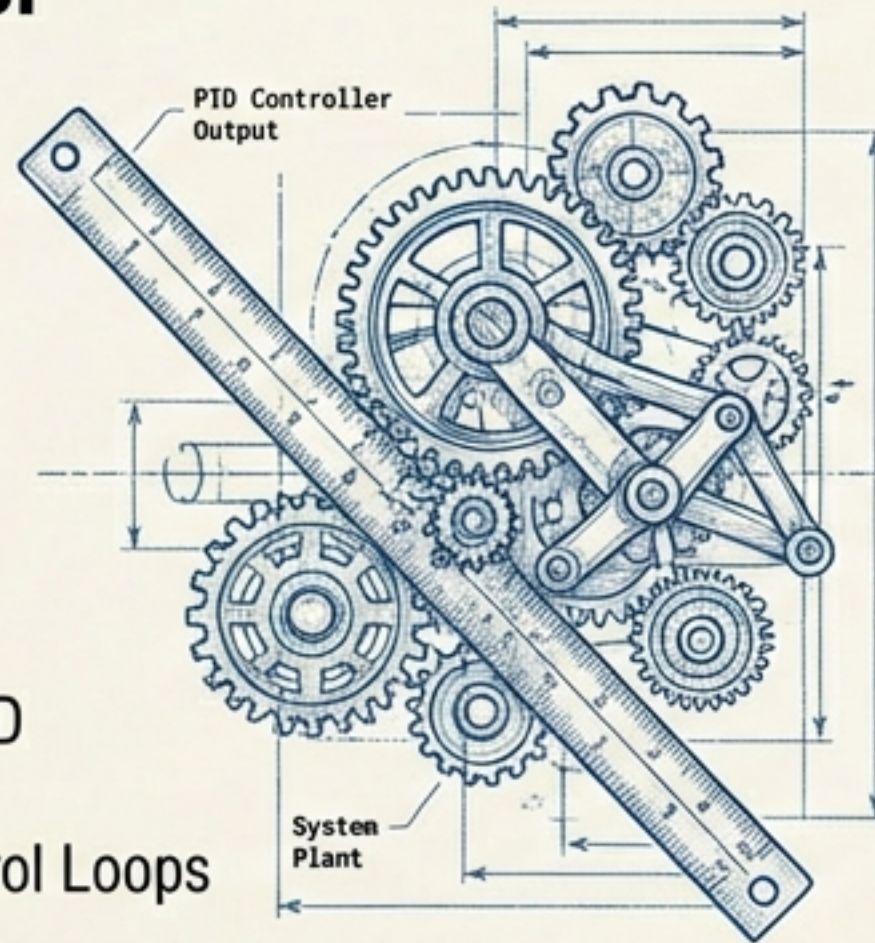
The Brain: From Calibration to Intuition

Replacing Linear PID with Deep Reinforcement Learning (DRL).

Traditional PID Control

Requires manual tuning.
Struggles with non-linear dynamics (wind, damage).
Separates perception from control.

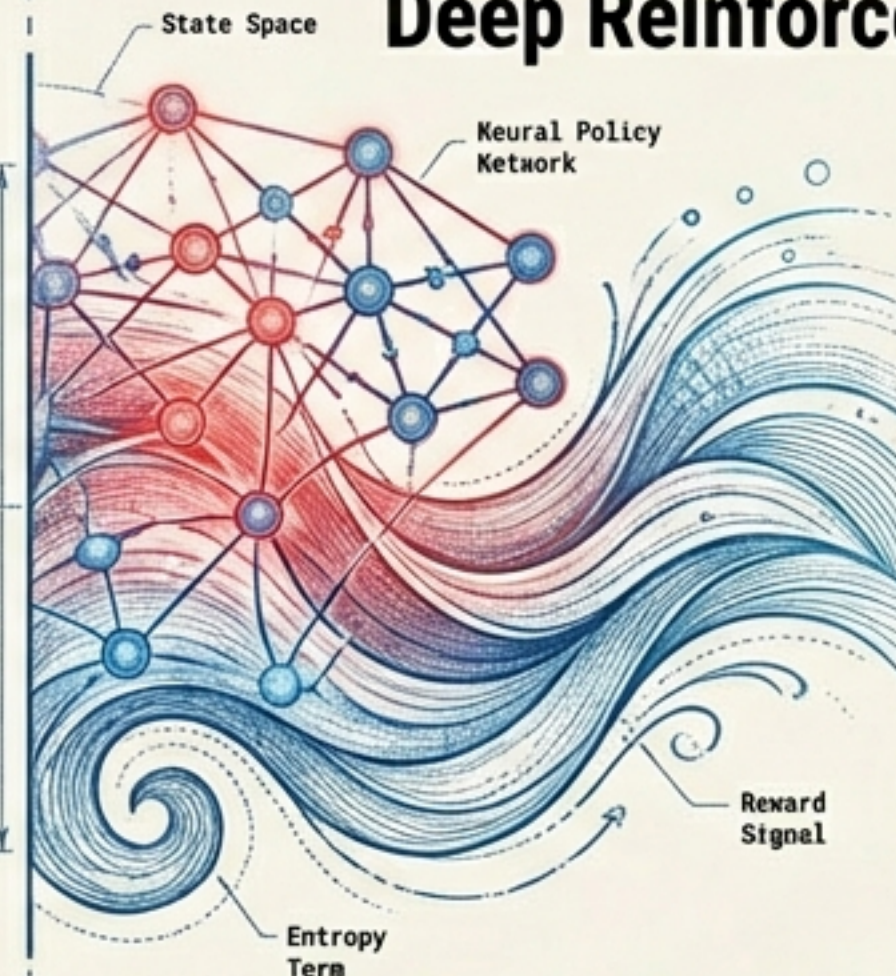
- Manual Gain Tuning: P, I, D
- Limited Adaptability
- Separate Perception/Control Loops



Deep Reinforcement Learning (DRL)

Learns control policies end-to-end. Handles complex coupled dynamics (roll + pitch).
Adaptable to chaos.

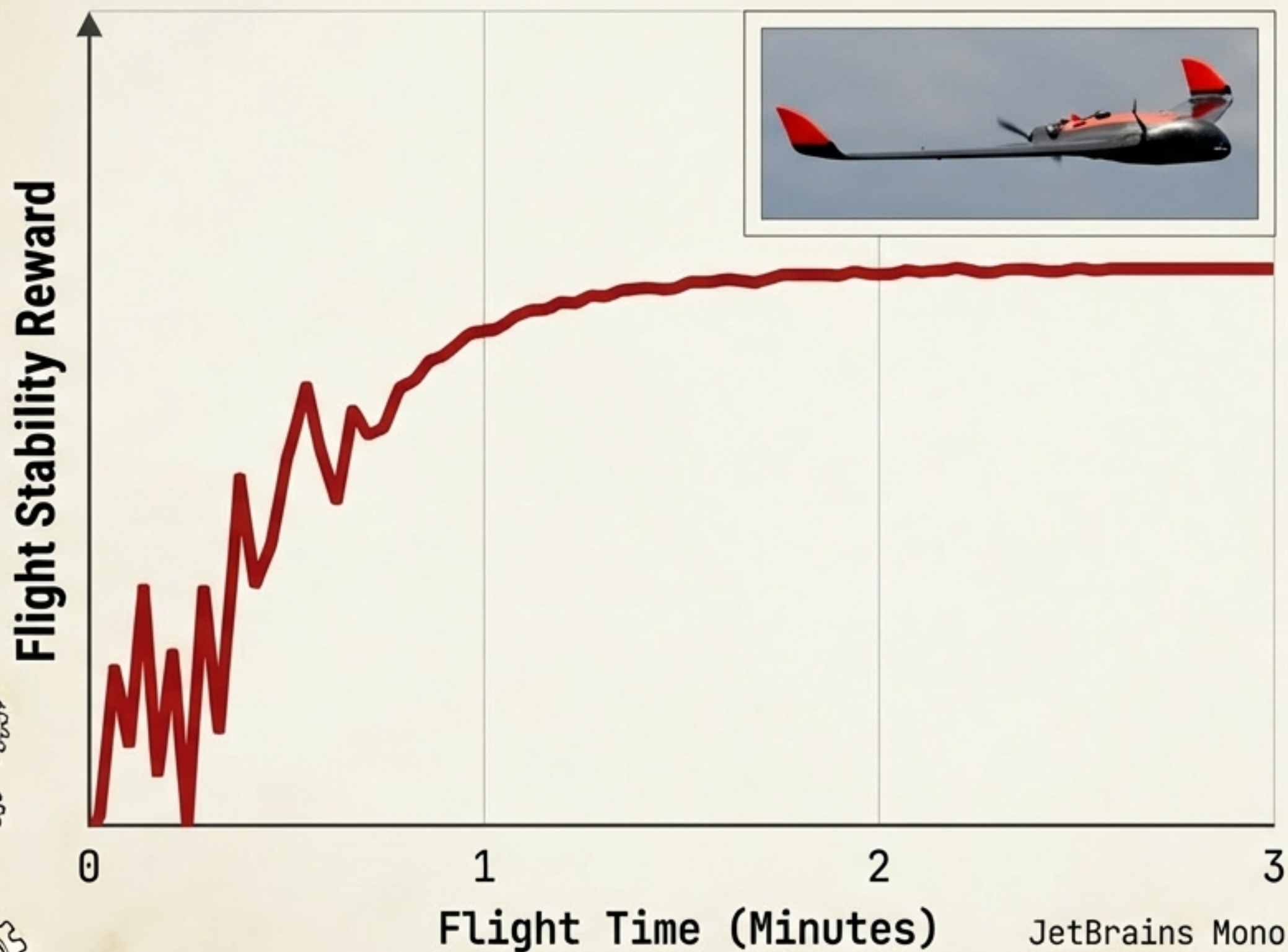
- End-to-End Learning
- Coupled Dynamics Handling
- Chaos Adaptable
- Continuous Learning



Method: Soft Actor-Critic (SAC)

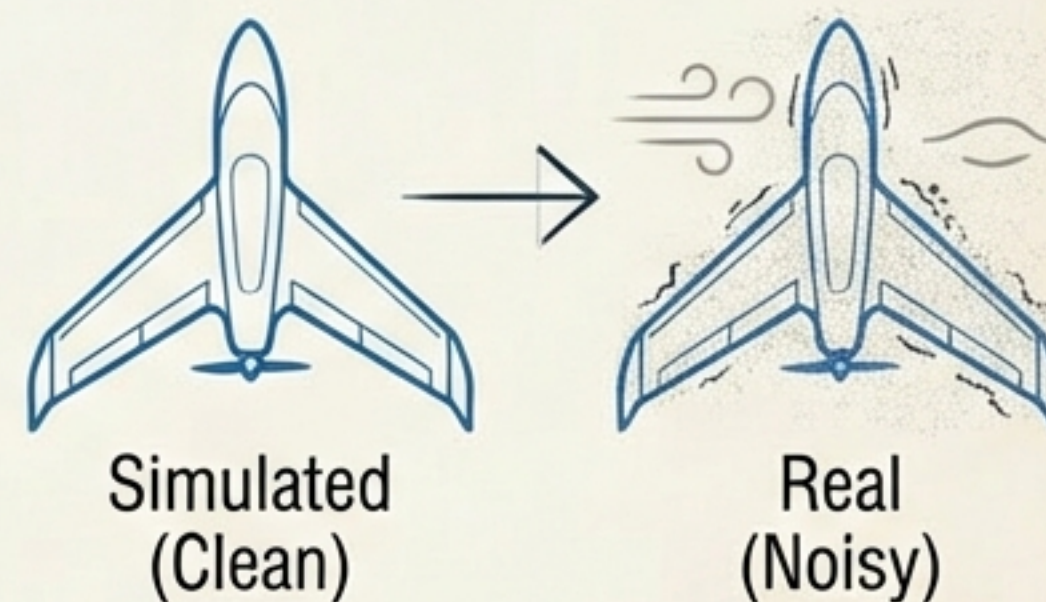
The agent optimizes a 'Sparse Reward' function. It gains points for stability and is penalized for aggressive, energy-wasting movements (CAPS). Entropy Regularization incentivizes safe exploration.

Learning to Fly in Three Minutes



Bridging the Reality Gap

Simulators are perfect; reality is messy. We use **Domain Randomization** during training to bridge this gap.

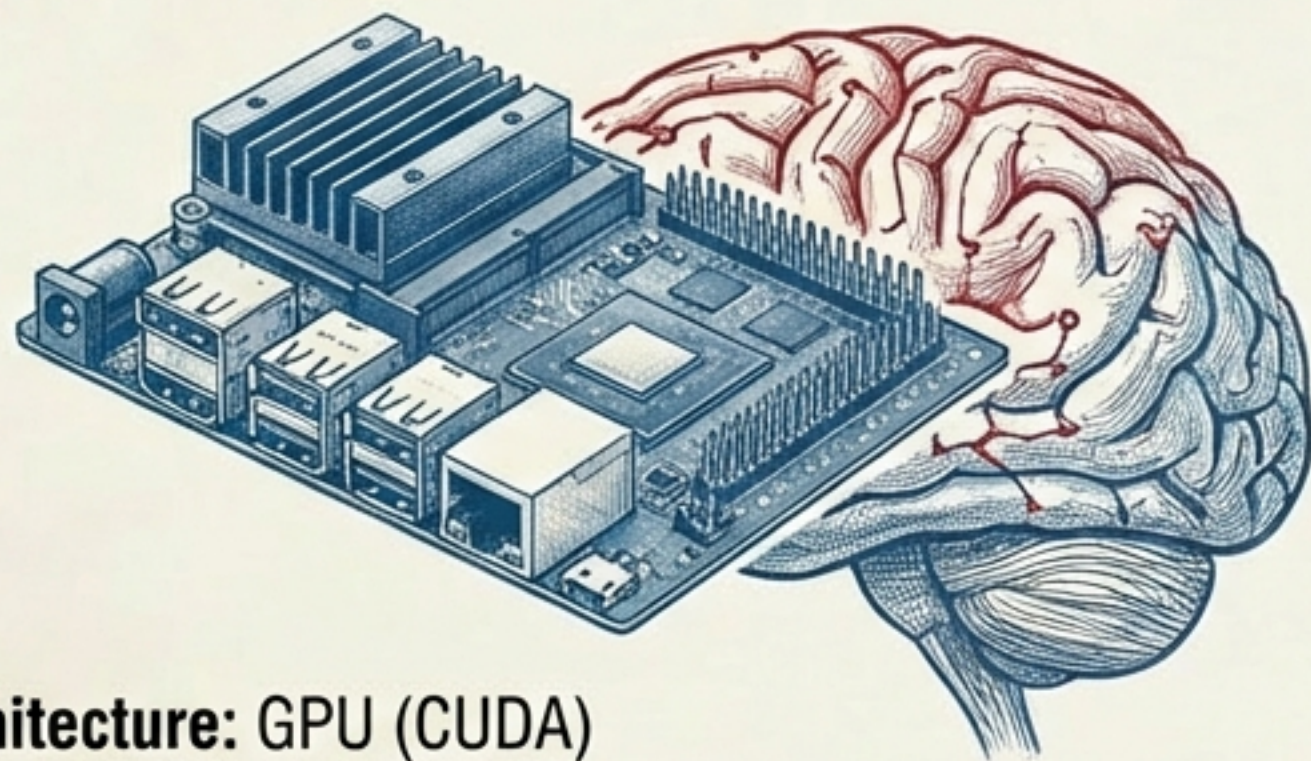


Note: Training with pessimistic latency (100ms) in simulation eliminates oscillations in real-world deployment.

The Cortex: Silicon Constraints

Balancing IQ (Compute) vs Metabolism (Energy).

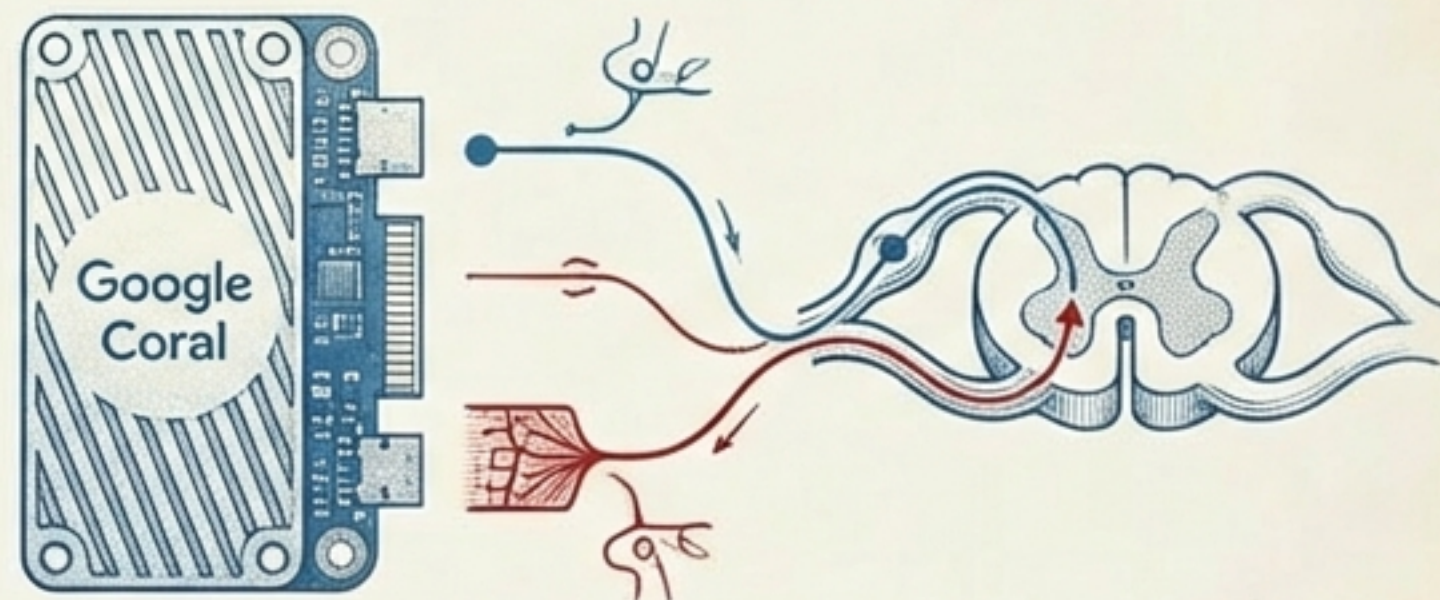
NVIDIA Jetson (The Generalist)



- **Architecture:** GPU (CUDA)
- **Power:** 5-15W (High Metabolism)
- **Best For:** Multi-model pipelines, video analytics.

The Frontal Lobe—powerful but hungry.

Google Coral TPU (The Specialist)

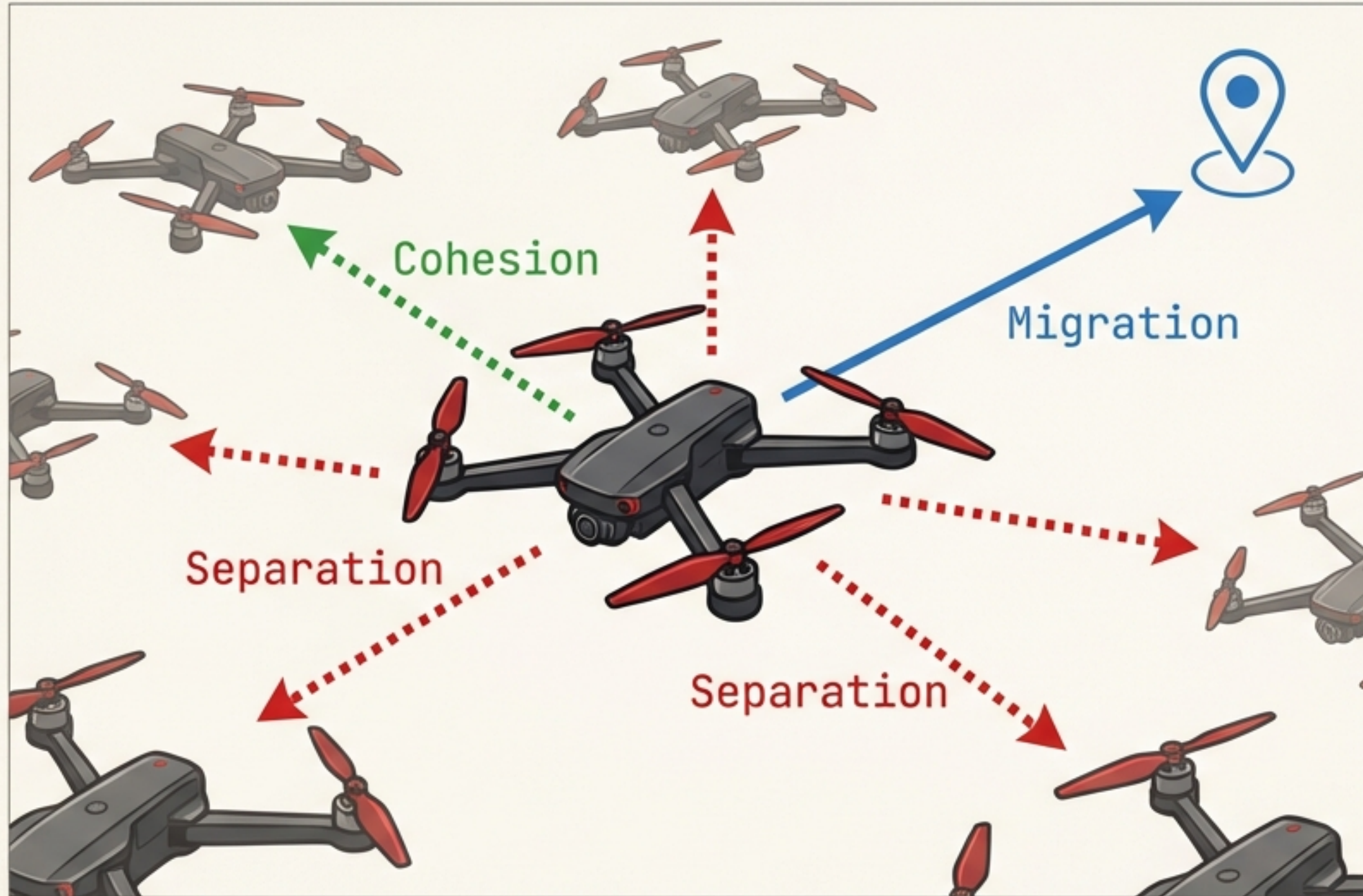


- **Architecture:** ASIC (TensorFlow Lite)
- **Power:** 2-4W (Ultra-Low Metabolism)
- **Best For:** High-speed inference, battery-critical missions.

The Reflex Arc—fast and efficient.

Decision Matrix: Choose Jetson for flexibility. Choose Coral for endurance.

The Society: Decentralized Cohesion



The Paradigm Shift

Moving from centralized command to **Vision-Based Flocking**. Each drone observes its neighbors to decide movement. **No GPS sharing. No radio communication.**

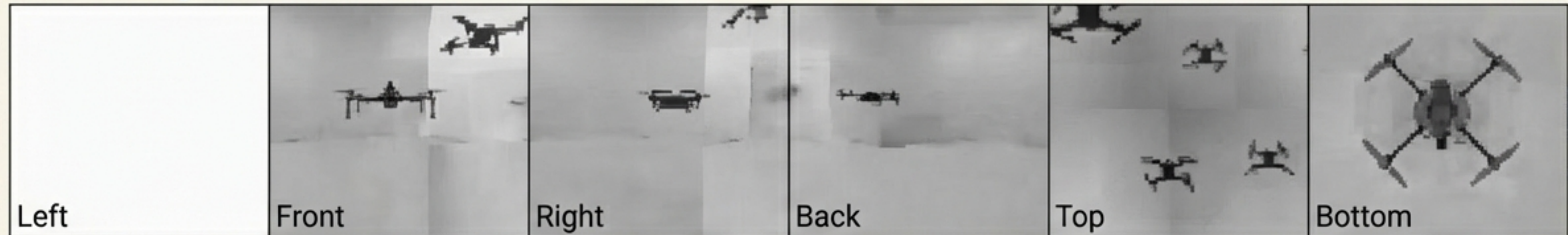
The Three Laws

1. Steer to avoid collision.
2. Steer toward the average center.
3. Steer toward the global objective.

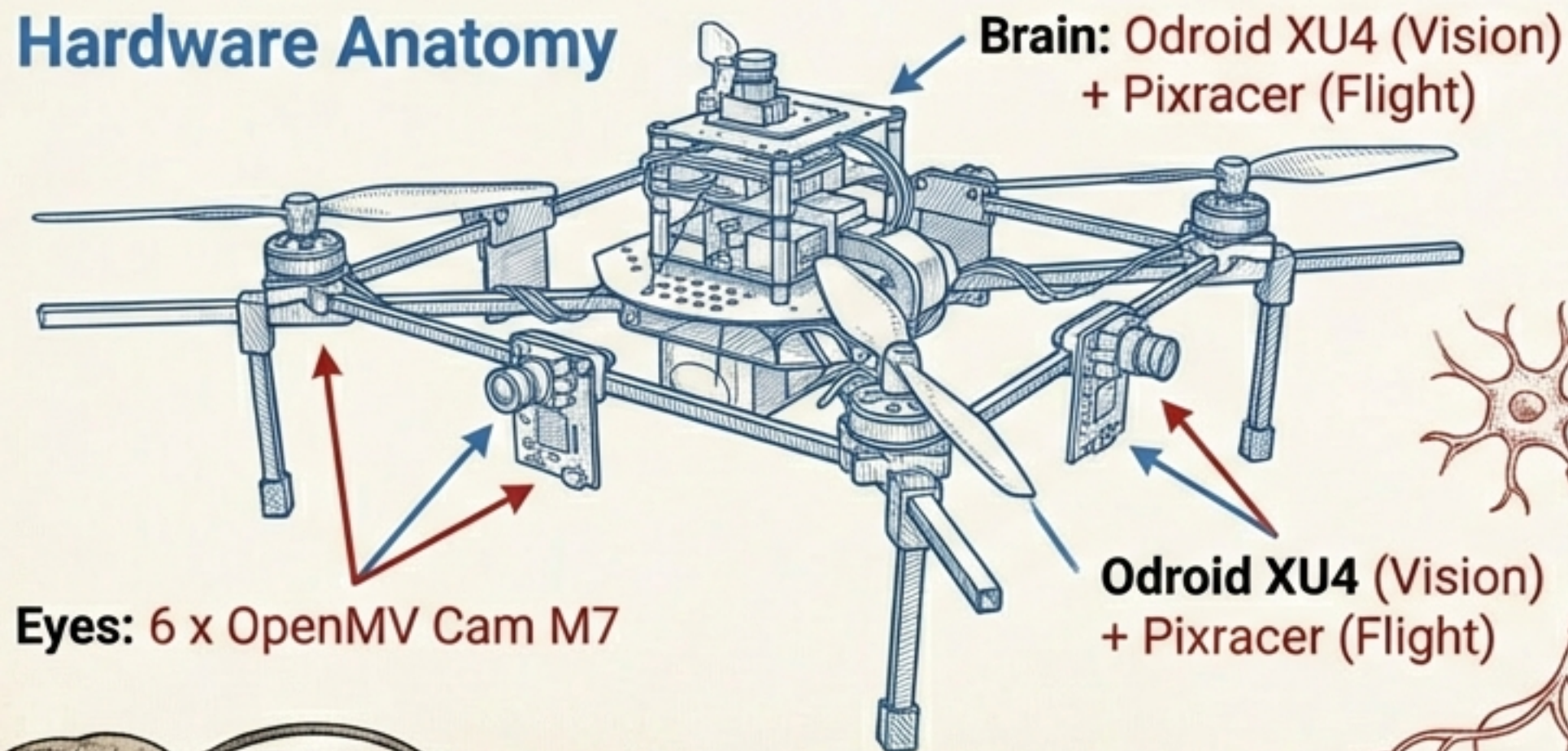
The Sensorium: Omnidirectional Awareness

Cube Map

360° Drone Sensor Feed



Hardware Anatomy



The Data Stream

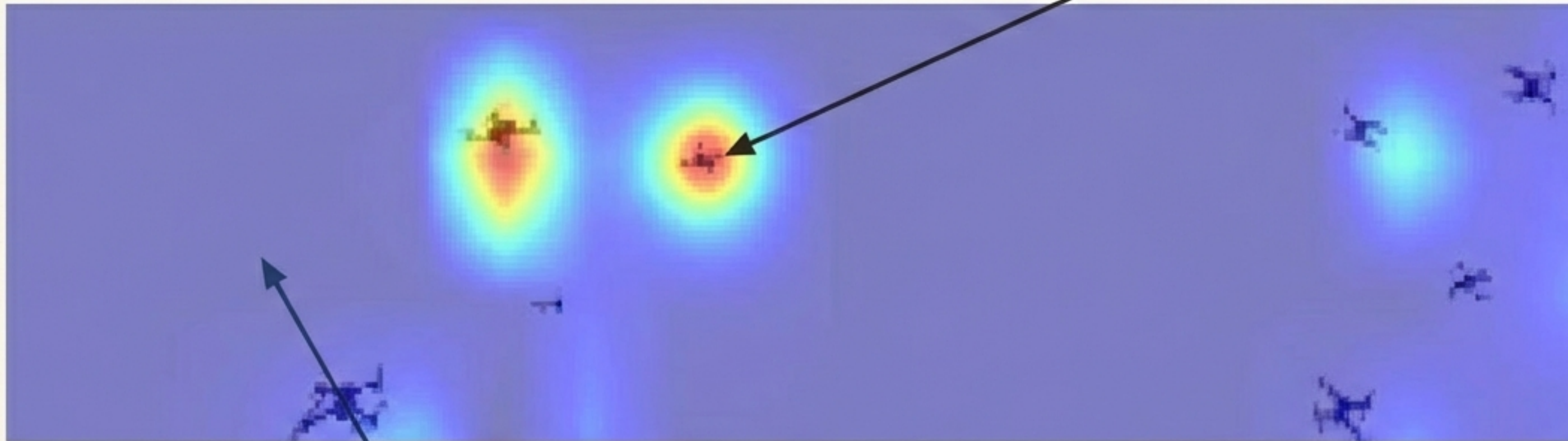
Raw pixel data is fed directly into the neural network to predict velocity commands (v_x , v_y , v_z).

Key Innovation

The controller learns relative positions from pixel density changes. No explicit coordinate lists are exchanged.

Attention Mechanisms: The Saliency Map

****Hot Zone:
Neighbor Detected****



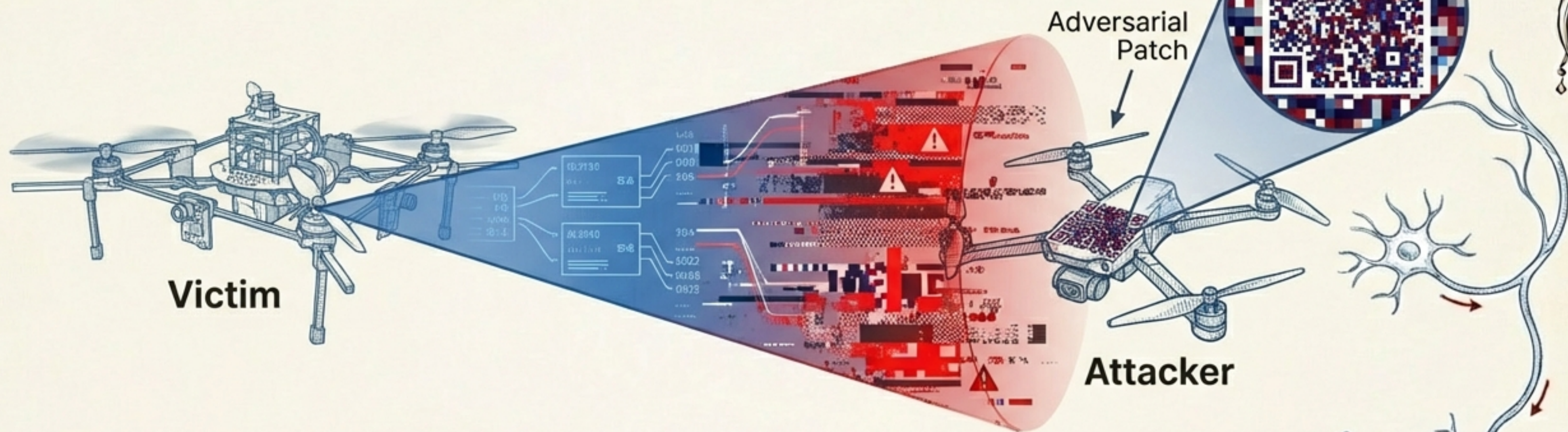
Cold Zone: Ignored

Insight

The neural network automatically learns to focus "attention" on neighboring agents to calculate separation/cohesion vectors. This detection capability emerged from training, without explicit object detection programming.

Vulnerabilities: The Attack Surface

White-Box Attacks and Adversarial Patches.



Visual Hacking:

If a drone relies on vision, it can be tricked. The neural network's interpretation of the environment is vulnerable to targeted manipulations.

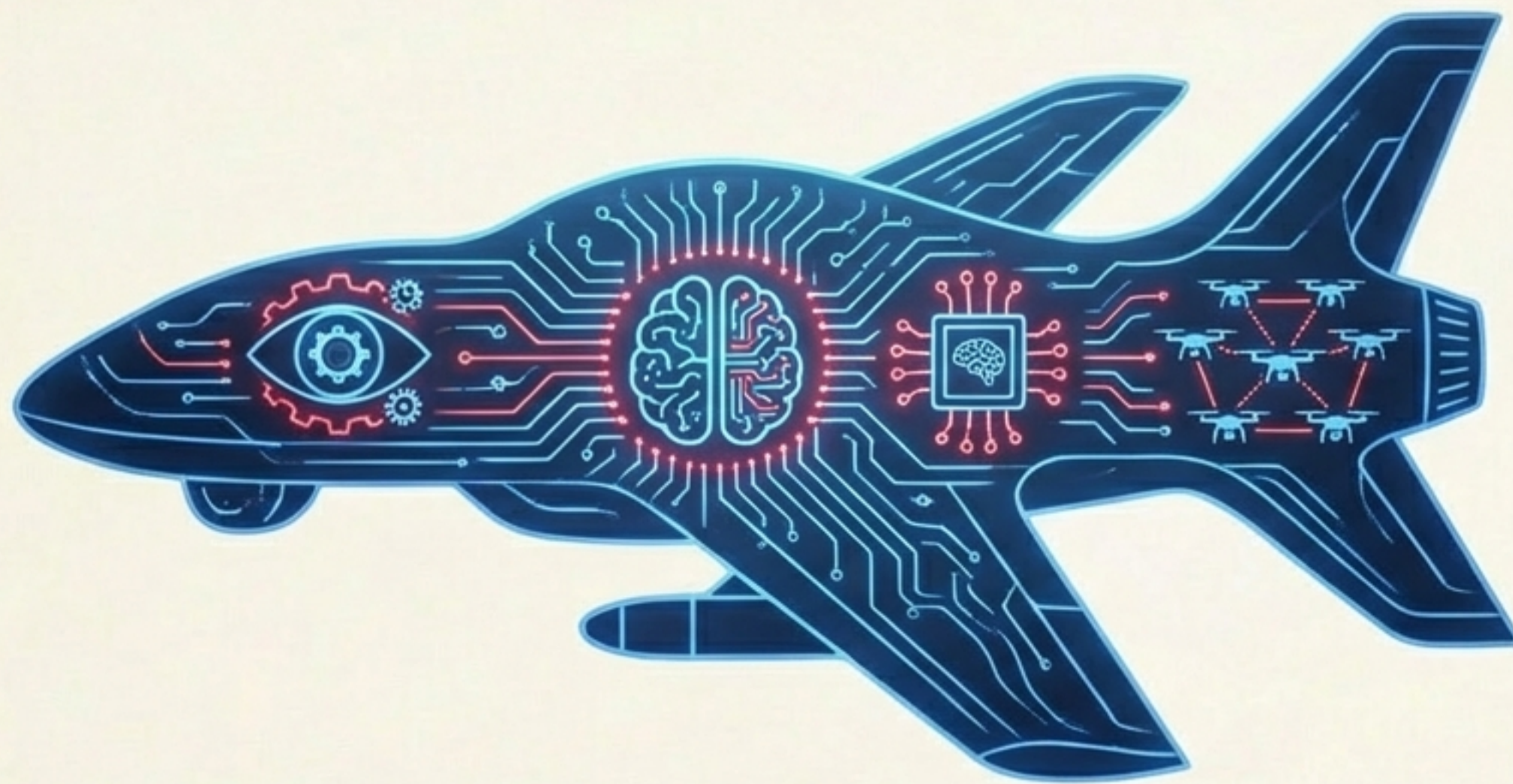
Adversarial Patches:

Specially printed patterns optimized to manipulate pixel values in a way that misleads the machine learning model. These physical stickers are designed to be interpreted as something else entirely, or to inject massive noise into the system.

The Consequence:

An attacker can hijack the victim's pose estimation, forcing it to deviate from its path, lose stability, or crash, simply by entering its field of view with a patch. The control system receives corrupted position and velocity data, leading to critical failure.

The Future: The Autonomous Agent



Eyes: UAD-YOLOv8
(High precision, low weight)

Brain: DRL/SAC
(Learned physics, robust to chaos)

Cortex: Edge TPU/GPU
(Watt-efficient processing)

Society:
Vision-based decentralization
(Robust to signal loss)

"We are no longer building flying cameras. We are building synthetic organisms capable of perception, decision, and cooperation in signal-denied environments. The tether is cut."